

Vydhra MERN + AI Job-Ready Bootcamp

Duration: 12 Weeks

Format: 12 Core Modules
+ AI

Level: Beginner to
Job-Ready Fullstack

3-Month Intensive Certification Roadmap

Module 01	Week 1 - Web Foundations	
Topics	HTML Semantic Structure, SEO Basics, Forms & Input Types	
	CSS Box Model, Flexbox, Tailwind CSS & Responsive Design	
Practical	Build a modern, responsive landing page with AI-assisted styling	
Outcome	Students can build accessible, mobile-first layouts using modern CSS frameworks.	

Module 02	Week 2 - JavaScript Mastery	
Topics	JS Essentials (ES6+), Variables, Functions, Arrays, Objects	
	Asynchronous JS, Promises, Async/Await & API Fetching	
Practical	Build an API-based mini productivity application	
Outcome	Students can handle complex data and asynchronous operations in JavaScript.	

Module 03	Week 3 - React Core	
Topics	React Basics, JSX, Components, Component Lifecycle	
	State Management, useEffect, Handling Events & React Router	
Practical	Build a multi-page interactive React user interface	
Outcome	Students can create dynamic single-page applications with seamless navigation.	

Module 04	Week 4 - Advanced React & State	
Topics	Context API, Global State, Authentication Flow	
	Protected Routes, Error Handling & UI Polish with AI	
Practical	Implement global state and secure authentication flows	
Outcome	Students can manage complex application states and secure UI components.	

Module 05	Week 5 - Frontend Production	■
Topics	Project Setup, Reusable Component Design, CRUD Integration	
	Frontend Optimization, UX Principles & Production Readiness	
Practical	Finalize a production-ready dashboard frontend with full CRUD	
Outcome	Students complete a professional-grade frontend system ready for hosting.	

Module 06	Week 6 - Node.js & Express	■
Topics	Node.js Fundamentals, Express Setup, Middleware Usage	
	REST API Design Principles & Service Layer Architecture	
Practical	Build a structured and scalable backend REST API	
Outcome	Students can build robust, modular server-side applications.	

Module 07	Week 7 - Databases & Prisma	■
Topics	Database Design (MongoDB/SQL), Prisma ORM & Schemas	
	Relational CRUD, Data Modeling & AI Query Optimization	
Practical	Design relational schemas and perform storage operations	
Outcome	Students can model complex data structures and optimize database performance.	

Module 08	Week 8 - Backend Security	■
Topics	JWT Authentication, Access/Refresh Tokens, Password Hashing	
	Role-Based Access Control (RBAC) & API Hardening	
Practical	Implement a secure login system with role-based permissions	
Outcome	Students can protect backend resources and secure sensitive data.	

Module 09	Week 9 - AI Tools & Workflows	■
Topics	Prompt Engineering, AI Code Generation, Testing with AI	
	Automated Documentation & AI-Driven Refactoring	
Practical	Use AI to automate tests, documentation, and code cleanup	
Outcome	Students adopt AI-first workflows for high-quality software development.	

Module 10	Week 10 - AI Integration	■
Topics	OpenAI API Integration, Chatbot Development	
	AI Blog Generators & AI Code Reviewer Assistants	
Practical	Build and integrate custom AI features into full-stack apps	
Outcome	Students can leverage LLMs to create intelligent application features.	

Module 11	Week 11 - Deployment & CI/CD	■
Topics	Environment Configs, Cloud Deployment (Vercel/DigitalOcean)	
	CI/CD Pipelines, Production Performance & Best Practices	
Practical	Deploy a full-stack system with automated CI/CD pipelines	
Outcome	Students understand the complete software release lifecycle in production.	

Module 12	Week 12 - Capstone & Portfolio	■
Topics	Capstone Development, Full-stack AI System Finalization	
	Resume, LinkedIn, GitHub Polish & Interview Prep	
Practical	Complete a portfolio-ready full-stack AI-integrated system	
Outcome	Students finish with a major project and job-ready career profiles.	